
PHYSICS

9702/22

Paper 2 AS Structured Questions

October/November 2016

MARK SCHEME

Maximum Mark: 60

Published

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Page 2	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – October/November 2016	9702	22

- 1 (a) (i) force/area (normal to the force) B1 [1]
- (ii) ($p = F/A$ so) units: $\text{kg m s}^{-2}/\text{m}^2 = \text{kg m}^{-1} \text{s}^{-2}$ A1 [1]
- allow use of other correct equations:
 e.g. ($\Delta p = \rho g \Delta h$ so) $\text{kg m}^{-3} \text{m s}^{-2} \text{m} = \text{kg m}^{-1} \text{s}^{-2}$
 e.g. ($p = W/\Delta V$ so) $\text{kg m s}^{-2} \text{m}/\text{m}^3 = \text{kg m}^{-1} \text{s}^{-2}$
- (b) units for m : kg, t : s and ρ : kg m^{-3} C1
- units of C : kg/s ($\text{kg m}^{-3} \text{kg m}^{-1} \text{s}^{-2}$)^{1/2}
 or
 units of C^2 : $\text{kg}^2/\text{s}^2 \text{kg m}^{-3} \text{kg m}^{-1} \text{s}^{-2}$ C1
- units of C : m^2 A1 [3]
- 2 (a) $\Delta E = mg\Delta h$ C1
- $= 0.030 \times 9.81 \times (-)0.31$
- $= (-)0.091 \text{ J}$ A1 [2]
- (b) $E = \frac{1}{2}mv^2$ C1
- (initial) $E = \frac{1}{2} \times 0.030 \times 1.3^2 (= 0.0254)$ C1
- $0.5 \times 0.030 \times v^2 = (0.5 \times 0.030 \times 1.3^2) + (0.030 \times 9.81 \times 0.31)$ so $v = 2.8 \text{ ms}^{-1}$
 or
 $0.5 \times 0.030 \times v^2 = (0.0254) + (0.091)$ so $v = 2.8 \text{ ms}^{-1}$ A1 [3]
- (c) (i) $0.096 = 0.030(v + 2.8)$ C1
- $v = 0.40 \text{ ms}^{-1}$ A1 [2]
- (ii) $F = \Delta p/(\Delta)t$ or $F = ma$
 $= 0.096/20 \times 10^{-3}$ or $0.030(0.40 + 2.8)/20 \times 10^{-3}$ C1
- $= 4.8 \text{ N}$ A1 [2]
- (d) kinetic energy (of ball and wall) decreases/changes/not conserved, so inelastic
 or
 (relative) speed of approach (of ball and wall) not equal to/greater than (relative) speed of separation, so inelastic. B1 [1]
- (e) force = work done/distance moved
 $= (0.091 - 0.076)/0.60$ C1
- $= 0.025 \text{ N}$ A1 [2]

Page 3	Mark Scheme	Syllabus	Paper
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- 3 (a) resultant force (in any direction) is zero B1
 resultant moment/torque (about any point) is zero B1 [2]
- (b) (i) force = $33 \sin 52^\circ$ or $33 \cos 38^\circ$
 = 26 N A1 [1]
- (ii) 26×0.30 or $W \times 0.20$ or 12×0.40 C1
 $26 \times 0.30 = (W \times 0.20) + (12 \times 0.40)$ C1
 $W = 15 \text{ N}$ A1 [3]
- (c) (i) $E = \Delta\sigma / \Delta\varepsilon$ or $E = \sigma / \varepsilon$ C1
 $\Delta\sigma = 2.0 \times 10^{11} \times 7.5 \times 10^{-4}$
 $= 1.5 \times 10^8 \text{ Pa}$ A1 [2]
- (ii) $\Delta\sigma = \Delta F / A$ or $\sigma = F / A$ C1
 $A = 78 / 1.5 \times 10^8 (= 5.2 \times 10^{-7} \text{ m}^2)$ C1
 $5.2 \times 10^{-7} = \pi d^2 / 4$
 $d = 8.1 \times 10^{-4} \text{ m}$ A1 [3]
- 4 (a) wave incident on/passes by or through an aperture/edge B1
 wave spreads (into geometrical shadow) B1 [2]
- (b) (i) waves (from slits) overlap (at point X) B1
 path difference (from slits to X) is zero/
 phase difference (between the two waves) is zero
 (so constructive interference gives bright fringe) B1 [2]
- (ii) difference in distances = $\lambda / 2 = 580 / 2$
 = 290 nm A1 [1]
- (iii) $\lambda = ax / D$ C1
 $D = [0.41 \times 10^{-3} \times (2 \times 2.0 \times 10^{-3})] / 580 \times 10^{-9}$ C1
 = 2.8 m A1 [3]
- (iv) same separation/fringe width/number of fringes
 bright fringe(s)/central bright fringe/(fringe at) X less bright
 dark fringe(s)/(fringe at) Y/(fringe at) Z brighter
 contrast between fringes decreases
- Any two of the above four points, 1 mark each B2 [2]

Page 5	Mark Scheme	Syllabus	Paper
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- 6 (a) hadron not a fundamental particle/lepton is fundamental particle
or
hadron made of quarks/lepton not made of quarks
or
strong force/interaction acts on hadrons/does not act on leptons B1 [1]
- (b) (i) ${}^0_1\text{e}^{(+)}$ or ${}^0_1\beta^{(+)}$ B1
- ${}^0_0\nu_{(\text{e})}$ B1 [2]
- (ii) weak (nuclear force / interaction) B1 [1]
- (iii) • mass-energy
• momentum
• proton number
• nucleon number
• charge
- Any three of the above quantities, 1 mark each* B3 [3]
- (c) (quark structure of proton is) up, up, down or uud B1
- up/u (quark charge) is $(+)\frac{2}{3}(\text{e})$, down/d (quark charge) is $-\frac{1}{3}(\text{e})$ C1
- $\frac{2}{3}\text{e} + \frac{2}{3}\text{e} - \frac{1}{3}\text{e} = (+)\text{e}$ A1 [3]